Original research

Inhalational exposures associated with risk of interstitial lung disease: a systematic review and meta-analysis

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ABSTRACT

Rationale Inhalational exposures are associated with risk of developing interstitial lung disease (ILD), yet the relationship between specific exposures and ILD is poorly characterised.

Objective Identify inhalational exposures associated with ILD and estimate the effects of exposures on ILD risk.

Methods MEDLINE and EMBASE databases were searched from 1990 to 2022 to identify inhalational exposures associated with ILD diagnosis. ILDs where causality is well-established (hypersensitivity pneumonitis, pneumoconiosis) and sarcoidosis were excluded. Two independent reviewers screened abstracts with full-text review and data extraction of eligible studies. Where possible, data were pooled and multilevel meta-analysis was specified using a random effects model. Sources of heterogeneity and risk of bias were assessed.

Main results 96 studies were included in the systematic review, representing 40 819 116 subjects (295 167 had ILD, 40 523 949 controls). For the meta-analysis, 54 studies were included (40 490 793 subjects: 273 899 ILD, 40 216 894 controls). Exposures associated with significantly increased ILD risk included smoking (OR 1.69, 95% CI 1.47 to 1.94), organic exposures (OR 1.56, 95% CI 1.12 to 2.16), metals (OR 1.52, 95% CI 1.07 to 2.16), dust (OR 1.45, 95% CI 1.20 to 1.76) and asbestos (OR 1.53, 95% CI 1.08 to 2.15). Silica and fumes had positive associations with ILD that trended towards significance.

Conclusions This systematic review and multilevel meta-analysis is the first to comprehensively assess the effect of inhalational exposures on overall risk of ILD, with multiple putative exposures identified. Future work should investigate novel occupational exposures associated with ILD, characterise the gene-environment interaction and develop preventative strategies.

PROSPERO registration number CRD42022292908.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Some specific disease-exposure relationships, such as smoking and idiopathic pulmonary fibrosis, have been found, but the magnitude and extent of the effect of inhalational exposures on interstitial lung disease (ILD) incidence remains incompletely understood.

WHAT THIS STUDY ADDS

⇒ This systematic review and meta-analysis found that smoking, organic exposures, metals, dust and asbestos were all significantly associated with an increased odds of ILD incidence.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Future research should prioritise the characterisation of specific dust exposures in ILD, mitigation of known occupational exposures and understanding of the role of inhalational triggers in early or subclinical ILD.

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INTRODUCTION

The interstitial lung diseases (ILDs) are a heterogeneous group of disorders characterised by inflammation and/or fibrosis of the pulmonary parenchyma. Incidence and prevalence of these disorders may be increasing, causing attendant rises in morbidity and early mortality.[1 2] While some forms of ILD are defined by the presence of ongoing or historical exposures (eg, organic exposures such as mould and hypersensitivity pneumonitis (HP); asbestos and pneumoconiosis), identification of inhaled exposures can also inform the diagnosis and management of idiopathic ILDs such as idiopathic pulmonary fibrosis (IPF).[3] For example, the IPF population attributable fraction (PAF, the proportion of cases that are due to an exposure) for occupational exposure to vapours, gases, dusts and fumes (VGDF) is 26%.[4] Approximately 45% of patients with IPF, connective tissue disease-related ILDs (CTD-ILDs) or unclassifiable ILDs reported exposure to organic, inorganic or both types of exposures in the previous 3 years.[5 6] In general, the strength and causal nature of these associations remain incompletely understood.[4]

Hundreds of potential ILD-related exposures have been identified over time, and the emergence of new technologies means that novel exposure-disease associations are recognised every year. For instance, being employed in a dental office, exposure to silica from artificial stone countertops, and indium from liquid-crystal display screens have all been putatively associated with ILD.[7–9] Some

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historical exposures may be outdated and others irrelevant, depending on the geographical region and individual patient. Given the vast number of potential exposures, clinicians cannot feasibly inquire about each individual one in a routine clinical encounter. Despite their obvious importance, there is a lack of guidance on how to ascertain ILD-related exposures, and of all possible exposures, which ones are most relevant. An evidence-informed contemporaneous and relevant list of aetiological exposures could improve the efficiency and accuracy of exposure ascertainment in both clinical and research settings.

The objective of this systematic review and meta-analysis was to comprehensively identify occupational and environmental exposures associated with increased risk of ILD, define the magnitude of risk and establish the relative importance of each exposure to inform future prioritisation of exposure assessments in clinical and research settings.

METHODS

The review was registered in PROSPERO (CRD42022292908) prior to commencement and conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (online supplemental table E1).[10]

Literature search and initial study assessment

A comprehensive search was conducted on two major databases: EMBASE Ovid and MEDLINE Ovid for English-language publications dated 1 January 1990 to 19 April 2022 (online supplemental tables E2 and E3). Search terms related to ILD and exposures were included in each search. Database subject headings (eg, MEDLINE Subject Heading), title abstract keywords and Boolean operators were used to define and combine these terms. We excluded pneumoconiosis (asbestosis, silicosis) in the search terms given the robust evidence of its causal relationship with exposure (online supplemental tables E2 and E3). We excluded HP given recent scoping reviews on specific causal exposures and the presence of a recently validated HP questionnaire.[11 12] Sarcoidosis was also excluded given the myriad pulmonary presentations of this disease that occur outside of ILD and the large volume of data on sarcoidosis that is outside the scope of our current objectives. No specific inhalational exposures were excluded. Eight previously identified references were used to test the search strategy to ensure it captured appropriate citations.[13–20]

Abstracts were uploaded to Covidence systematic review management software (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia, available at www.covidence.org) to facilitate screening processes. All titles and abstracts were independently screened for eligibility for full review by two reviewers (CL and SE). Full review eligibility criteria were applied as follows. Participants were adult (age ≥18 years) humans with ILD, including IPF, idiopathic interstitial pneumonia, CTD-ILD, unclassifiable ILD, interstitial pneumonia with autoimmune features and smoking-related ILDs or controls without ILD. Excluded ILDs were HP, sarcoidosis, asbestosis, silicosis, other pneumoconiosis, drug-related ILD and radiation-induced ILD. The intervention/exposure was defined as any inhaled environmental, occupational, domestic or avocational exposure(s). The comparator/control was defined as those not exposed to any inhaled environmental, occupational, domestic or avocational exposures, or, in descriptions where the entire cohort is exposed, those without ILD. The primary outcome was ILD diagnosis, with the secondary outcome of a relative ranking of risk estimates for ILD diagnosis, and the tertiary outcome of a comparison of risk associations across diagnostic subtypes. Additional inclusion requirements were publication in the English language and use of original data to report an association between exposure and risk of ILD. Case reports, case series with fewer than 10 patients, and publications without a numerical effect estimate for the association of exposure to the risk of ILD were excluded.

Full-text review was completed by two reviewers (CL and SE) following screening. Discrepancies at each stage of screening (ie, title and abstracts and full-text screening) between reviewers were resolved through discussion with a third reviewer (KAJ).

Full-text review and data extraction

For publications qualifying for full-text review and extraction, extracted study characteristics included study title, year of publication, study authors, design, cohort name, location, sample size (proportion of cases vs controls). Cigarette smoking history including smoking status (former, current, never) and estimated smoking dose (eg, number of pack years of smoking) was extracted. Data on exposures including exposure type (eg, mould, metal), method of assessment (eg, from clinical history/chart review, self-report), location of exposure (eg, occupational, residential) were extracted. The outcome of interest was the diagnosis of an ILD as specified above. If reported, the specific ILD subtypes were recorded along with diagnostic method: administrative (eg, International Classification of Diseases (ICD) codes), clinical (eg, multidisciplinary diagnosis), radiological or other. In studies where the effect estimate of the same exposure was reported across multiple subgroups (ie, smoking in men and smoking in women, or current smoking and former smoking), these were reported as unique effect estimates (ie, one study was allowed to have multiple effect estimates within the same exposure grouping). If exposures were tested across multiple methods in the same paper; however (ie, self-report and job-exposure matrix), priority was given to exposures that were examined via job exposure matrix or expert designation. Similarly, when both unadjusted and adjusted ORs were reported, only adjusted ORs were extracted. Effect estimates and 95% CIs, their type (eg, OR, HR) and analytical covariates, if reported, were also extracted, along with sex, age and race. Studies that did not have this effect estimate but had some description of inhalational exposures in relation to ILD incidence were not meta-analysed but included in the systematic review.

Study quality assessment

Risk of bias was performed using the Newcastle-Ottawa Quality Assessment Scale.[21] This assessment was only performed on studies meeting criteria for meta-analysis. Each included study was assigned a score between 0 and 7 indicating their risk of bias as per the tool's three quality domains: selection, comparability and outcome. We modified this scale in the comparability domain by assigning a star if adjustment was made for smoking or any other type of coexposure. We adjusted the outcome domain by assigning a star if a multidisciplinary diagnosis of ILD or an administrative diagnosis of ILD using ICD-9 codes was obtained; we also assigned a star for follow-up if the prospective studies had a minimum duration of 10 years of follow-up and automatically assigned this star for the retrospective studies. Studies were deemed of 'good' quality if they were given over five stars, 'fair' quality if between three and five stars and poor if less than three stars. As a sensitivity analysis, Agency for Health Research and Quality (AHRQ) definitions of quality were

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applied to each study's Newcastle-Ottawa Assessment score, consistent with other literature.[22]

Statistical analysis

Descriptive statistics were used to characterise the parent studies. A qualitative and quantitative synthesis of the association between the various reported exposures and ILD was conducted. To enable a pooled effect estimate for similar exposures that were named differently across studies (eg, 'rock dust' in one study and 'cement dust' in another), specific exposures were grouped into the following categories: smoking, air pollution, silica, asbestos, organic exposures, metals, vapours, gases, dusts, fumes, and the latter four also grouped into one category, VGDF. Designations into categories were determined via consensus of three authors (CL, SE, KAJ).

Although the majority of estimates were reported as ORs, when different types of effect estimates were reported, risk/HRs and their respective 95% CIs were converted to ORs based on accepted equations (online supplemental figure E1). Multilevel meta-analysis (MLMA) was specified with random effects modelled using restricted maximum likelihood method. Log ORs were included as effect estimates and specified along with their SEs. Due to the grouping of specific exposures into categories, individual studies included multiple estimates. Random effects were specified at the level of individual study-specific exposure and the individual study itself. The MLMA model was specified with adjustment for exposure and diagnostic group (IPF, CTD, organising pneumonia (OP) and all other diagnoses were classified as other ILD). Marginal effects were estimated to visualise overall and stratified effect estimates by exposure. As a sensitivity analysis, traditional meta-analysis with pooled effect estimates using Der Simonian and Laird random effects models is included in the supplement. Metaregression was also performed to assess the effect of study year on the most commonly studied exposure (smoking). Heterogeneity was assessed according to the I^2 statistics at each level and combined, using the Higgins-Thompson method and Cochran method. All analyses were performed using Stata (StataCorp, 2022, Release 18).

RESULTS

Search results and study characteristics

The literature search identified 10 100 studies (figure 1). After duplicates were removed, 8056 abstracts were screened and 295 studies underwent full-text review. This yielded 96 studies for inclusion in the systematic review and/or meta-analysis, representing a total of 40 819 116 subjects (online supplemental table E4). Of these, 295 167 had a diagnosis of ILD, and 40 523 949 were non-diseased controls. 38 studies were from Europe, 32 were from North America, 20 were from Asia and 6 were from other continents. Most of the studies (n=90) were published after the year 2000. Nearly all studies were of retrospective design: 49 studied retrospective cohorts, 36 were case-controls and 5 were cross-sectional. Six studies examined a prospectively collected cohort. 54 studies had effect estimates of exposure on the risk of ILD diagnosis and were thus included in the meta-analysis; 42 studies did not meet meta-analysis criteria but were included in the systematic review. Of the studies included in the meta-analysis, 22 studied retrospective cohorts, 29 were case-control studies, 1 was cross-sectional and 3 studied a prospectively collected cohort.

Using star number as criteria, study quality was good in 11 studies, fair in 39 and poor in 4 studies (table 1). Using the more stringent AHRQ criteria, study quality was good in 11 studies, fair in 12 and poor in 31 studies.

Exposures and associations with ILD diagnosis: systematic review

Smoking was the most common exposure reported, assessed in 69 studies (25 984 237 total subjects). Various ILDs were associated with smoking in these studies, including IPF (22 studies), rheumatoid arthritis (RA)-ILD (22 studies), other CTD-ILDs (6 studies) and other ILDs (19 studies). Air pollution and its constitutive elements were investigated for ILD associations in five studies, two in CTD-ILD and three in IPF. Asbestos exposure was examined in six studies, three with the outcome of IPF, one with the outcome of cryptogenic fibrosing alveolitis and two with the outcome of other ILDs. Silica was associated with non-pneumoconiosis ILD in four studies; two examining scleroderma-ILD, another with the outcome of IPF, one study with the outcome of OP and two with other ILDs. Additional studies that investigated specific occupational exposures but did not reach the standards required for the meta-analysis described associations with firefighting, hairdressing, World Trade Center dust exposure, cement dust exposure, textile dye printing and oil impregnation.

While 84 studies included self-reported exposures, 8 had a job-exposure matrix or other validation of exposure via an occupational medicine specialist. Additionally, four studies of air pollution used geocoded residential addresses linked to ambient pollution measurement stations, and four studies used either serological or pathological identification of specific particulates in patient serum or tissue respectively.

MLMA analysis of ILD diagnosis by exposure

54 studies were included in the meta-analysis (40 490 793 subjects: 273 899 ILD, 40 523 949 controls). The exposure groupings associated with a significantly increased risk of ILD overall were smoking (OR 1.69, 95% CI 1.47 to 1.94), organic exposures (OR 1.56, 95% CI 1.12 to 2.16), metals (OR 1.52, 95% CI 1.07 to 2.16), dust (OR 1.45, 95% CI 1.20 to 1.76) and asbestos (OR 1.53, 95% CI 1.08 to 2.15) (figure 2). Notably, air pollution and gases were not associated with increased odds of ILD in this meta-analysis, while silica and fumes had ORs that trended towards significance. Results were not substantially different in the traditional meta-analysis performed as a sensitivity analysis (online supplemental table E5). In an additional sensitivity analysis excluding the nine studies that did not report any adjusted effect estimates, all in the smoking subgroup, the risk of ILD with smoking was identical (OR 1.69, 95% CI 1.47 to 1.94).

When stratified by ILD diagnosis, smoking was associated with increased odds of IPF (OR 1.65, 95% CI 1.38 to 1.98), CTD-ILD (OR 1.74, 95% CI 1.33 to 2.26) and other ILD (OR 1.72, 95% CI 1.37 to 2.14) (figure 3). Silica was associated with increased odds of other ILD (OR 1.61, 95% CI 1.01 to 2.50), although its relationship to both IPF and CTD trended toward significance. Organic exposure was associated with increased odds of both IPF (OR 1.54, 95% CI 1.09 to 2.18) and other ILD (OR 1.60, 95% CI 1.13 to 2.27), and exposure to metals was associated with increased odds of IPF (OR 1.50, 95% CI 1.03 to 2.19) and other ILD (OR 1.56, 95% CI 1.08 to 2.26). Exposure to fumes was associated with IPF (OR 1.50, 95% CI 1.05 to 2.14) and other ILD (OR 1.56, 95% CI 1.08 to 2.25), and dust exposure was associated with increased odds of IPF (OR 1.44, 95% CI 1.16 to 1.79) and other ILD (pooled OR 1.50,

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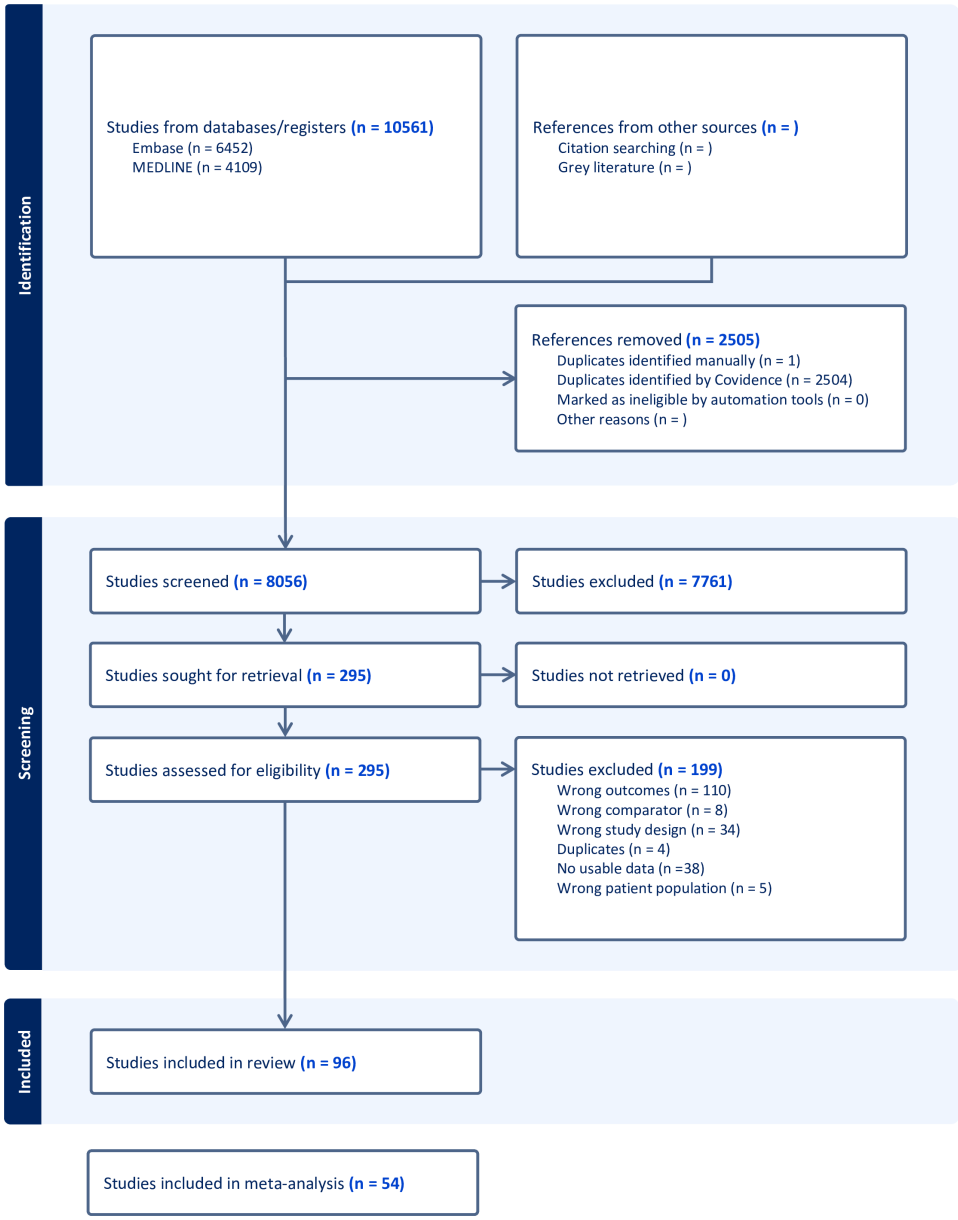

Figure 1 PRISMA (preferred reporting items for systematic reviews and meta-analyses) diagram.

95% CI 1.17 to 1.92). Asbestos was significantly associated with IPF (OR 1.50, 95% CI 1.04 to 2.16) and other ILD (OR 1.56, 95% CI 1.08 to 2.26). Metaregression did not demonstrate any significant effect of study year on odds of ILD (online supplemental table E6). Total study heterogeneity was high (I^2>50%) in all estimates via the Higgins-Thompson method and high in all except fumes and metals in the Cochran method (online supplemental table E6).

Between-study heterogeneity was higher than within-study heterogeneity for most estimates, with I^2 of 96.5 (between) and 1.7 (within) for air pollution, 59.4 (between) and 10.8 (within) for dust, 61.5 (between) and 0 (within) for fumes, 98.8 (between) and 0.7 (within) for metals and 56.8 (between) and 0 (within) for organic subgroups (online supplemental table E6).

DISCUSSION

This systematic review and meta-analysis is the first to comprehensively assess the effect of inhalational exposures on the risk of ILD as a broad category of lung disease. We found multiple exposures were consistently associated with increased odds of ILD diagnosis, including smoking and dust, and workplace exposures such as asbestos, metal and potentially silica. These associations were consistent across ILD subtypes, emphasising the effects of inhalational exposures across all forms of ILD. These findings highlight the importance of exposure ascertainment for all patients with ILD and the need for improved characterisation of inhalational risk and impact of mitigation. Other systematic reviews and meta-analyses have focused on subsets of exposures, such as occupational exposures or air pollution, or subsets of disease, such as IPF.[23–25] This is the first analysis of all known exposures in conjunction with the risk of a broad range of ILD diagnoses. This work addresses an important knowledge gap, particularly given the evolving recognition of shared biology across ILD subtypes.[26]

When comparing the strength of associations between exposures and ILDs, smoking and organic exposures had the highest pooled OR of ILD, followed by metals, dust and asbestos, with silica and fumes trending towards significance. Misclassification

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Table 1 Risk of bias assessment of studies included in the meta-analysis

	Selection				Comparability	Outcome		
Author, Year	Representativeness	Selection of unexposed	Ascertainment of exposure	Demonstration of outcome	Key variables	Assessment of outcome	Follow-up adequacy	Total
Abramson *et al* 2020[13]	*	*	*	*	*	*	*	7
[39]Andersson 2021	*	*	*	–	*	*	*	6
Bae 2022[40]	*	*	–	–	*	*	NA	4
Baumgartner *et al* 2000[19]	*	–	*	–	*	–	*	4
Bellou 2021[41]	–	*	–	–	*	*	–	3
Ben Takaya 2022[42]	–	–	–	*	–	*	*	3
Berge 2003[43]	–	–	*	–	*	–	NA	2
Chen 2020[44]	*	*	*	–	–	*	*	5
Choi 2018[45]	*	*	–	*	*	*	–	5
Cramer 2016[46]	*	*	–	–	–	*	*	4
Cui 2023[47]	–	*	*	–	*	*	–	4
Diaz de Leon 2010[48]	–	*	–	–	–	*	*	3
Ekstrom 2014[49]	*	–	–	–	*	–	*	3
Fang 2024[50]	–	–	–	*	*	*	*	4
Figueroa 2010[51]	*	*	*	–	*	*	*	6
Garcı'a-Sancho 2011[51]	*	*	*	–	*	*	*	6
Gribbin 2008[52]	*	*	–	–	–	–	*	3
Gustafson *et al* 2007[17]	*	–	–	–	*	–	*	3
Harari 2019[53]	*	*	–	–	–	*	–	3
Hubbard *et al* 1996[15]	*	*	–	–	*	*	*	5
Iversen 2023[54]	*	*	*	–	*	*	–	5
Jobard 2017[55]	*	*	*	–	*	*	*	6
Kaul 2022[56]	*	*	–	–	*	*	*	5
Kaul 2022[57]	*	*	–	–	*	*	*	5
Kawabata 2015[58]	–	*	–	–	*	*	*	4
Kim 2017[59]	*	*	–	–	*	–	*	4
Kim 2023[60]	*	*	–	*	*	–	*	5
Kitamura 2007[61]	–	–	*	*	*	*	*	5
Konak 2024[62]	*	*	–	*	*	*	*	6
Koo 2017[63]	*	*	*	–	*	*	*	6
Kronzer 2021[64]	*	*	–	*	*	*	*	6
Lamichhane 2019[65]	–	*	–	–	–	*	*	3
Lilleker 2017[66]	*	–	*	*	–	*	*	5
McDermott 2024[67]	*	*	–	*	*	*	*	6
Mena-Vasquez 2021[68]	*	*	–	–	*	*	*	5
Miyake 2005[69]	*	–	–	–	*	*	*	4
Mori 2012[70]	*	*	–	*	*	–	*	5
Mullen 1998[71]	–	–	*	–	–	–	*	2
Muntyanu 2022[72]	–	*	–	–	*	–	*	3
Paolocci *et al* 2018[20]	*	*	–	–	*	–	*	4
Park 2024[73]	*	*	–	–	*	*	*	5
Paulin 2021[74]	*	–	–	–	–	–	*	2
Pugh 2016[75]	*	*	*	*	*	*	*	7
Ren 2023[76]	*	*	–	*	*	–	*	5
Restrepo 2015[77]	*	*	–	–	NA	–	–	2
Reynolds 2022[78]	*	–	*	–	*	*	*	5

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Table 1 Continued

Author, Year	Selection				Comparability	Outcome		
	Representativeness	Selection of unexposed	Ascertainment of exposure	Demonstration of outcome	Key variables	Assessment of outcome	Follow-up adequacy	Total
Salvatore 2015[79]	–	*	–	*	*	–	*	4
Schiffenbauer 2018[80]	*	*	*	*	*	*	*	7
Scott *et al* 1990[14]	–	*	–	–	*	*	*	4
Steele 2005[81]	–	*	–	*	*	*	*	5
Wang 2018[82]	*	*	–	*	*	–	*	5
Wang 2015[83]	*	*	–	*	*	–	*	5
Wang 2023[84]	–	*	*	–	*	*	–	4
Zheng 2022[85]	*	*	–	–	*	*	*	5

could have occurred (eg, silicosis and asbestosis mistakenly diagnosed as other ILDs), especially as asbestosis and IPF can be indistinguishable. However, as myriad radiological patterns can arise from the same inhalational exposure, the association between these exposures and other ILDs remains plausible.[27] While silica is well known to cause silicosis, a form of pneumoconiosis that was excluded from this analysis, its association with scleroderma is also well described.[28] Asbestos and silica are predominantly occupational exposures, and thus our data suggest that clinicians should prioritise the occupational history and potential attendant exposures when assessing patients for risk of ILD. Furthermore, these findings emphasise the importance of assessing for inorganic exposures in the clinical assessment of ILD risk, not merely the organic exposures that are classically associated with HP.

The relationship between smoking and ILD, particularly smoking and IPF, is well established and underscored by our study findings.[29] While the mechanism underpinning this relationship is not entirely understood, cigarette smoke has been found to activate collagen synthesis and fibroblast proliferation via TGF-β1 signalling in mouse models of pulmonary fibrosis.[30]

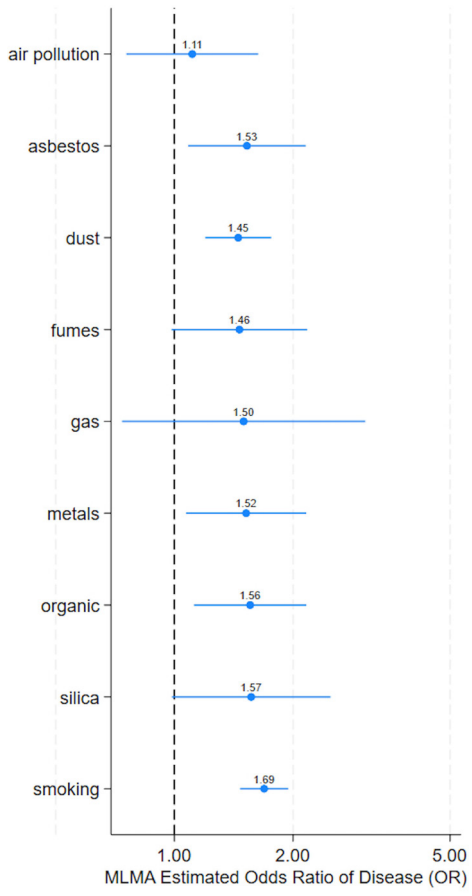

Figure 2 Multilevel meta-analysis of OR of ILD incidence by exposure group. ORs greater than 1 (more rightward) indicate a significant relationship between exposure and disease incidence. ILD, interstitial lung disease; MLMA, multilevel meta-analysis.

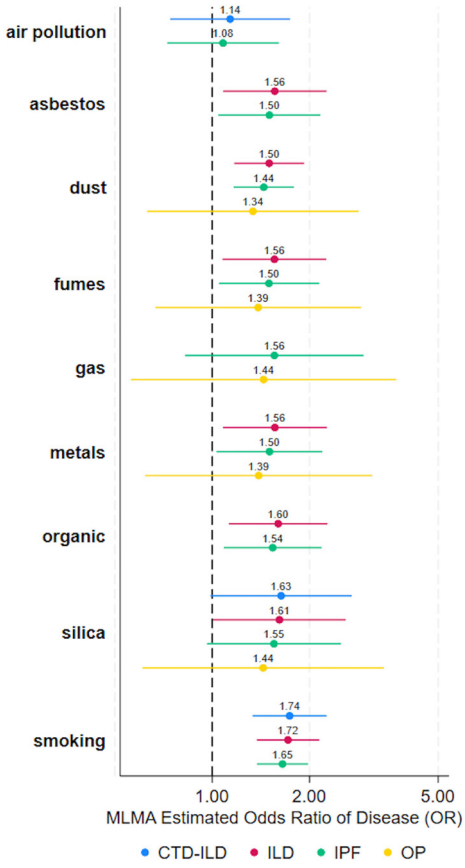

Figure 3 Multilevel meta-analysis of OR of ILD incidence by exposure group, stratified by ILD diagnosis. ORs greater than 1 (more rightward) indicate a significant relationship between exposure and disease incidence. CTD-ILD, connective tissue disease-interstitial lung disease, ILD, other ILD, IPF, idiopathic pulmonary fibrosis; MLMA, multilevel meta-analysis; OP, organising pneumonia.

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We found this effect of smoking on ILD incidence was present in both IPF and non-IPF ILDs, which is relatively novel and historically under-appreciated outside of the smoking-related ILDs. This finding illustrates the need to refine exposure assessment tools, even for those with other causes for ILD like CTD.

VGDF, a well-described entity encompassing exposure risk in patients with COPD, is receiving increased attention in the ILD literature. Most notably, a systematic review found high PAFs of occupational VGDF in common ILDs, including IPF, HP and sarcoidosis.[4] However, VGDF is relatively nonspecific, and our meta-analysis found differential ILD risk based on its constituent components. While no studies included here analysed vapours individually, dusts were consistently associated with ILD risk, fumes had borderline significance and gases showed no significant relationship. These differences may be due to variation in the size of particulates, as smaller diameter particles are more likely to be deposited in the lung parenchyma.[31] Interestingly, when stratified by diagnosis, dust was significantly associated with IPF as well as other ILDs. This finding is consistent with previous reports suggesting that inhalational exposures can trigger ILDs considered to be of known aetiology.[32] Future studies should clarify if the relationship between dusts and non-IPF ILD is universal or found only within specific disease subtypes.

The lack of association between air pollution exposure and ILD risk in our study may be reflective of the heterogeneous impacts of air pollution and its constitutive components; our meta-analysis did not contain enough effect estimates of individual pollutants to warrant meta-analysis by specific pollutant type. Additionally, the long latency period between exposure and disease incidence in ILD makes any causative associations with remote pollution exposure difficult to determine, while temporally closer outcomes such as ILD progression or near-term lung function abnormalities have consistently been tied to increases in air pollution.[33] Finally, one study of patients with RA did show an association with air pollution and ILD, but this study was unable to be meta-analysed due to a lack of sufficient reported data to convert HRs to ORs.[34] Our overall findings are not unexpected given an anticipated bias towards the null when air pollution is considered as a single exposure. Ongoing work in this field is focused on the most impactful components, namely $PM_{2.5}$ and ultrafine particulates, which will continue to inform pathobiology and risk.[35]

Strengths of this study include its search across multiple databases, double review to mitigate bias and comprehensive assessment of most categories of ILDs without limitation to one individual subtype. This is especially important as the field shifts from a historical paradigm of subtype-guided ILD diagnosis towards grouping by disease behaviours, including the phenotypes and endotypes shared by progressive, fibrotic disease.[36] Limitations include high heterogeneity for many of the estimates, imperfect grouping of individual exposure types into groups and high risk of bias for many studies, which is inherent given no validated measure of exposure exists along with the long latency of disease. A potential additional limitation contributed by case-control studies is healthy participant bias, by which differential participation by cases and controls in a study can skew effect estimates.[37 38] As all studies included were observational in nature, the amplitude of individual effect estimates should be interpreted with caution. The self-reported nature of many of these exposures, while somewhat unavoidable, is an additional source of potential bias. We excluded the pneumoconioses and HP from this analysis due to the clear causal relationship between exposure and disease. We also excluded sarcoidosis due to the sheer volume of data in this field, as well as the heterogeneity of pulmonary presentations not always clearly defined as ILD. The objective of this study was to characterise relationships between exposures and ILD subtypes not already clearly associated with inhalational exposures; however, these exclusions may limit the scope of our findings.

In conclusion, this systematic review and meta-analysis found many inhalational exposures associated with disease risk across the ILD spectrum. Future research should prioritise the characterisation of dust exposures in ILD, mitigation of known occupational exposures and understanding of the role of inhalational triggers in early or subclinical ILD.

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